FIELD NOTE

PROJECT WEEKLY REPORT - W2 - 0223-0227

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FIELD NOTE: SWEDISH ELECTRICITY PRICE FORECASTING

Period: Week 2 (Feb 23 – Feb 27, 2026)
Scope: Deep Data Cleaning, Core Feature Engineering, and Pipeline Verification

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1. EXECUTIVE SUMMARY

This week focused on transitioning from raw data acquisition to a production-ready feature engineering pipeline. The primary objective was to implement a robust `DataProcessor` capable of handling the inherent noise and gaps in multi-source energy data. Key milestones include the deployment of a deep cleaning module for timezone and outlier management, and the engineering of core power-specific features (Rolling Averages, Volatility, and Momentum) that are critical for capturing market dynamics in Sweden's SE3 bidding zone.

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2. COMPLETED TASKS

A. Deep Data Cleaning & Normalization

1. **Timezone Unification**: Integrated `convert_to_swedish_time` into the pipeline to ensure all data points align with `Europe/Stockholm` (CET/CEST), resolving DST transition overlaps.
2. **Continuous Time Indexing**: Implemented `reindex` logic to identify and fill hourly gaps, ensuring a gap-free timeline essential for lag feature calculations.
3. **Missing Value Imputation**: Deployed linear interpolation for target variables (Price/Load) to maintain temporal continuity.
4. **Extreme Outlier Management**: Implemented an IQR-based capping strategy (3.0 * IQR) to handle price spikes without introducing data leakage or artificial noise.

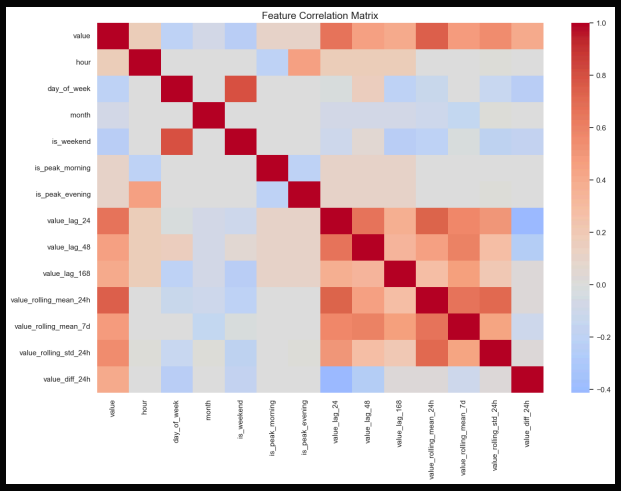
B. Core Feature Engineering (`src/features.py`)

1. **Temporal Features**: Added `hour`, `day_of_week`, `month`, and `is_weekend` flags.
2. **Peak Hour Indicators**: Defined specific features for morning (07:00-10:00) and evening (17:00-20:00) price peaks.
3. **Advanced Power Features**:
   - **Rolling Mean (24h, 7d)**: Captures short-term and weekly trends.
   - **Volatility (24h Std)**: Quantifies market NERVOUSNESS and price volatility.
   - ∗Momentum (24h Diff)**: Measures sudden price shifts relative to the previous day.
4. **Lag Features**: Generated 24h, 48h, and 168h lags for historical look-back.

C. Verification & Diagnostic EDA

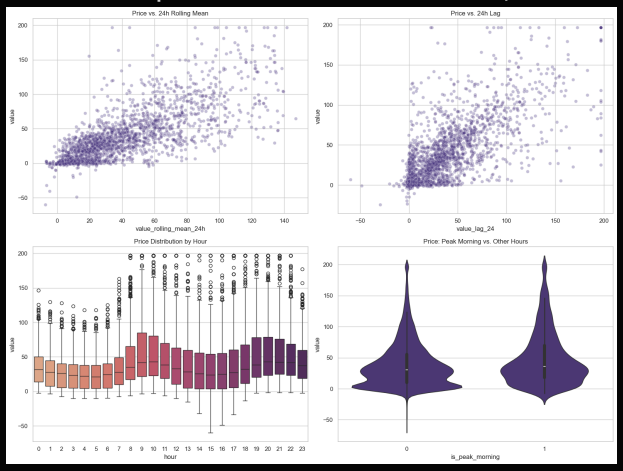
1. **Pipeline Validation Script**: Created `scripts/verify_features.py` to automate NaN detection, shape verification, and basic correlation checks.
2. **Feature Analysis Notebook**: Developed `02_EDA_and_Visualizations.ipynb` (refined) and `03_Feature_Engineering_Analysis.ipynb` to visualize feature-target relationships.

D. Feature Engineering Insights (Analysis of Notebook 03)



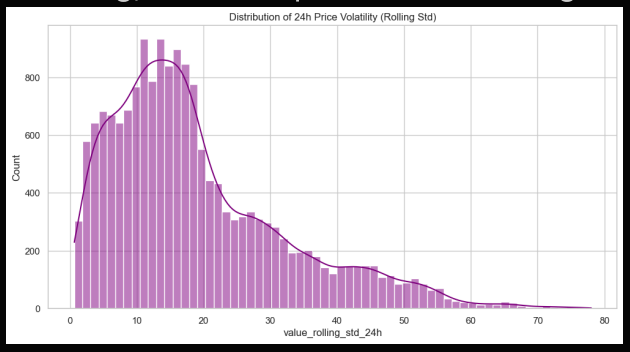
05_freature_correlation_matrix_0228.png

1. **Feature Correlation Matrix**: Confirmed `value_rolling_mean_24h` and `value_lag_24` as dominant predictors (corr > 0.75), validating the market's short-term dependency.



06_price_rlling_mean_0228.png

2. **Target vs. Key Features**: Scatter plots validated the linear relationship after deep cleaning, while boxplots confirmed rigid morning peak (07:00-10:00) patterns.



07_disruibution_of_24h_price_valatility_0228.png

3. **Volatility Distribution**: 24h rolling std showed a clean "long-tail" distribution, confirming that the IQR capping strategy effectively balanced signal retention and noise reduction.

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3. TECHNICAL CHALLENGES & SOLUTIONS

Challenge	Impact	Engineering Solution
Boundary NaNs	Rolling/Lag features create NaNs at the start of series.	Implemented `fillna(method='bfill')` to maintain consistent DataFrame shapes for training.
Price Volatility	Extreme spikes distort statistical distributions.	Applied **Capping (Clip)** instead of removal to preserve temporal sequence.
DST Ambiguity	Duplicate/Missing hours during clock shifts.	Used `ambiguous='infer'` in `tz_localize` within the core utility library.

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4. DEVELOPMENT PROGRESS & CODE STATUS

- **Logic**: `DataProcessor` class in `src/features.py` is fully operational and tested.
- **Verification**: `scripts/verify_features.py` confirms 0 NaNs and high feature relevance (e.g., 24h Mean corr: 0.75).
- **Data**: SE3 processed features are saved to `data/processed/se3_features_v1.parquet`.

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5. PLANS FOR NEXT WEEK (WEEK 3)

- [ ] **SMHI Weather Finalization**: Fully integrate temperature and wind speed into the `integrate_external_data` method.

- [ ] **Feature Selection (VIF)**: Perform Variance Inflation Factor analysis to prune highly collinear features.
- [ ] **Model Baseline**: Train initial XGBoost and LightGBM models to establish a performance baseline for SE3 price forecasting.
- [ ] **Hyperparameter Strategy**: Design a cross-validation framework for time-series data.